Exploring Sentiment and Emotion Trends Across Movies and Genres in IMDb User-Generated Reviews

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1. Introduction

1.1. Motivation Behind the Study

IMDb (Internet Movie Database) holds a large number of reviews and ratings generated by viewers and cinephiles. Understanding this type of audience's reactions and preferences to films using computational methods reveals valuable patterns about the role of media in shaping sentimental and emotional experiences. This research investigates how sentiments and emotions appear in movie reviews through advanced natural language processing (NLP) methods at both the individual film and, more importantly, the impactful genre levels.

The analysis of audience reactions using computational methods produces significant practical applications for filmmakers together with marketers, streaming platforms, and recommendation systems. Consider, for example, that instead of just looking at traditional star ratings that often obscure nuanced emotional responses, customer decision-making could be significantly improved by looking at the overall sentiment/emotion in reviews. The latter plays a crucial role in every film's success or, generally, the sales of a product. Building on this idea, Topal and Ozsoyoglu (2016) claim that "moviegoers" or "reviewers," as they call them, inherit an emotional state that contradicts the emotions a movie transmits to the audience. As a result, highly rated movies' evoked emotions may not match a viewer's anticipated emotional state. To further explore these ideas, this research study uses a mixed-methods approach—a combination of quantitative sentiment/emotion modeling with qualitative linguistic analysis—with a goal to achieve a detailed understanding of cinematic discourse.

1.2. Research Questions and Significance

This study attempts to shed focus on testing if genre influences sentiment/emotion patterns independently of individual movies. This study addresses the following two major research questions generated by the overall research aim, which tend to act as the driving force throughout this piece of academic writing:

RQ1: How are sentiments and emotions distributed across individual movies on IMDb user reviews, and how reliable are these distributions based on model confidence scores?
RQ2: Are different emotions (joy, fear, anger, etc.) dominant across different movie genres?

2. Related Work

2.1. IMDb as a Platform for Movie Reviews

The paper by Talibzade (2023) adds to the use of IMDb as the most popular source of information on movies, TV shows, and other media content because it provides real user ratings and reviews. The author argues that further research is required for conducting sentiment analysis on IMDb movie reviews in order to gain valuable insights for various stakeholders in the film industry. Therefore, it can be useful to understand audience preferences through sentiment analysis in order to enhance the quality of marketing strategies and film productions.

Gunawan, Alhafidh, and Wahyudi (2022) assert that the 1-10 rating range hides important users' opinions and emotional taste, because viewers are very rarely extreme about their ratings (a perfect score of 10). Instead, the majority occupy the space between extremes. This diverse rating choices indicate that the audience expresses their thoughts about different film elements, which results in complex emotions that mere numeric scores fail to properly represent.

2.2.Sentiment and Emotion Analysis of Movie Reviews

Since the 2000s, sentiment analysis (SA) has emerged as a widely studied field within natural language processing. The field has expanded its development for more than twenty years, which demonstrates its rising value for understanding text-based emotions and opinions. In light of this, the number of published papers on sentiment analysis demonstrated a gradual increase from 2002 until 2021, with rapid growth starting from 2015 and reaching its peak in 2021 as the year of most publications. This suggests that sentiment analysis has gained increasing popularity because of technological progress and its growing usage in multiple real-life applications (Cahyo et al., 2023; Cui et al., 2023).

A study by Talibzade (2023) utilized a dataset of 50,000 IMDb movie reviews, which were preprocessed and vectorized using different techniques. In addition, this paper delivers an extensive evaluation of the difference between traditional machine learning approaches and transformer-based models, specifically BERT, for sentiment analysis of IMDb movie reviews. The main goal was to select the optimal sentiment classification method for movie reviews, taking into consideration the accuracy, time efficiency, and computational resource

requirements. Traditional machine learning techniques produced acceptable sentiment classification results through Logistic Regression (90% accuracy), SVM (90% accuracy), and Naïve Bayes (MNB) with TF-IDF vectorization reaching 87% accuracy. On the other hand, the BERT model produced outstanding accuracy results, which reached 98% in sentiment classification. Based on this high accuracy result, transformer-based models provide superior performance compared to traditional approaches for this specific task. However, the BERT model delivered better accuracy results but needed additional computational power and a longer training duration than traditional machine learning techniques. Therefore, sentiment analysis model selection requires practitioners to evaluate the tradeoff between model performance and resource requirements.

There have been similar findings, as confirmed by Paneru, Thapa, and Paneru (2025) with a dataset of 50k IMDB reviews. The model used in this paper is a hybrid deep learning model that combines RoBERTa embeddings with a Convolutional Neural Network (CNN), referred to as the R-CNN model. The latter reached a test accuracy of 91.5%, which exceeded traditional methods, demonstrating its effectiveness in sentiment analysis. The authors highlighted the development of a Flask-based application that shows how their proposed R-CNN model works in real-time sentiment prediction. Users can enter movie reviews through the application which generates sentiment predictions to demonstrate how the R-CNN model effectively analyzes sentiment.

As a means to highlight gaps throughout the course of this review of literature, it was clearly identified that there's still a need for more research on how various movie genres might influence the specific sentiments expressed by viewers, as there was a focus on sentiment analysis without delving deeply into genre-specific emotional trends. This emphasizes the clear contribution of this piece of academic paper in exploring the mentioned area and building on existing literature.

3. Data Collection

3.1. Dataset Description

This research utilizes a dataset of nearly 1 million distinct movie reviews stemming from 1150 IMDb movies distributed across 17 different genres (Pal, Barigidad, & Mustafi, 2020). The data also includes various metadata per genre about the selected movies, such as release dates, IMDb ratings, and total review count per movie. The dataset is available under a Creative Commons Attribution license, which is acceptable for academic research.

For the first research question which examines sentiment and emotion distributions across individual movies and evaluates the models used through confidence scores, four representative movies spanning different genres were selected, as illustrated in Table 1. For each movie, the total number of reviews varies since this part does not directly compare movies.

Movie	**Year**	**IMDb Rating**	**Main Genre**	**Number of Analyzed Reviews**
Deadpool	2016	8/10	Comedy	1438 reviews
The Conjuring	2013	7.5/10	Horror	2082 reviews
The Mist	2007	7.1/10	Cosmic horror	1184 reviews
The Green Mile	1999	8.6/10	Drama	3334 reviews

Table 1: Metadata for Representative Movies Analyzed for Sentiment and Emotion Distributions

For the second research question, this study centers its focus on sentiment and emotion analysis across movie genres. Five primary genres—Drama, Thriller, Horror, Comedy, and Romance—were chosen from the original dataset of 17 genres based on their diversity and

commonness. A subset of reviews was sampled for each genre with 5,120 maximum reviews for each genre to ensure computational feasibility while maintaining representativeness. The aim of this selection technique is to successfully and effectively investigate sentimental and emotional patterns specific to a particular genre.

3.2. Text Preprocessing Pipeline

The dataset of the collected IMDb movie reviews went through a comprehensive text normalization process for the sake of data preparation for RoBERTa-based sentiment and emotion analysis. First, HTML tags were removed using regex pattern matching. Noise was dealt with by eliminating URLs, special characters, and whitespace to single-spaces normalization. Additionally, only English-language reviews were kept to maintain language consistency using langdetect. Finally, the dataset was validated to test that it was clean, and the original dataset was reduced to 662,310 reviews.

4. Methodology

This section outlines the models, tools, and processes employed in sentiment and emotion analysis.

4.1. Tools and Models

This research paper leverages two pretrained transformer models. For sentiment classification, the cardiffnlp/twitter-roberta-base-sentiment-latest model was employed. Using the TweetEval benchmark, this RoBERTa-based model, developed by Cardiff University's NLP Research Group, is trained on approximately 124 million tweets and more accurately detects sentiments—positive, neutral, negative. For this research, this model is ideal because it performs satisfactorily well, particularly in analyzing short, informal texts.

The SamLowe/roberta-base-go_emotions model, released in 2022, was applied for emotion analysis. It was trained on the GoEmotions dataset, which is a Reddit dataset annotated with 28 distinct emotion labels, which supports multi-label classification of emotions. This model enables inference at scale and offers top-k emotion predictions with corresponding scores.

4.2. Batch Processing

For performance optimization and data analysis in a reduced amount of time, the analysis of the extensive dataset required efficient processing through batched review evaluation. The HuggingFace Transformers library processed each batch through the sentiment and emotion pipelines. The approach provided scalability while maintaining computational efficiency. The batch processing system enabled parallel inference operations, preserving data integrity and producing aggregated results.

5. Findings and Evaluation

5.1. Sentiment and Emotion Distributions Across Movies

This subsection addresses RQ1: How are sentiments and emotions distributed across individual movies on IMDb user reviews, and how reliable are these distributions based on model confidence scores?

Metric	Average Confidence (0-1)
Sentiment Analysis	0.75
Emotion Detection	0.66

Table 2: Overall Model Confidence Scores

As shown in Table 2, the reliability of sentiment analysis and emotion detection results was ensured by calculating average confidence scores for the used models. While sentiment analysis shows higher confidence, emotion detection was noisier and reflects the inherent challenge of determining nuanced emotional expressions in user reviews. These scores recognize the variation in emotional classification while confirming the findings' validity.

The results revealed distinct sentiment distributions across movies, as Figure 1 illustrates:

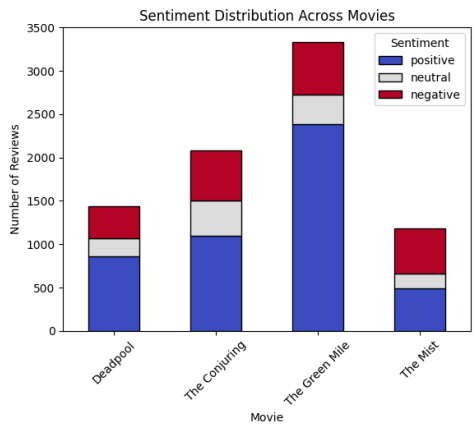

Figure 1: Sentiment Distribution Across Selected Movies

At 71.4%, The Green Mile (1999) dramatic narrative's emotional impact on the audience strongly aligns with the positive sentiment that dominated the results. However, The Mist (2007) displayed an unusual sentiment distribution, with negative sentiment (44.3%) slightly exceeding positive sentiment (41.4%), particularly reflecting viewers divided opinions on the film's controversial ending. This film's sentiment classification contradicts its high rating score on IMDb (7.5/10), which highlights important revelations that understanding reviews nuanced sentiments could be far more reliable than ratings in improving audience decision-making.

Emotion analysis explores in more depth the specific feelings conveyed by reviewers, as demonstrated by the following figures:

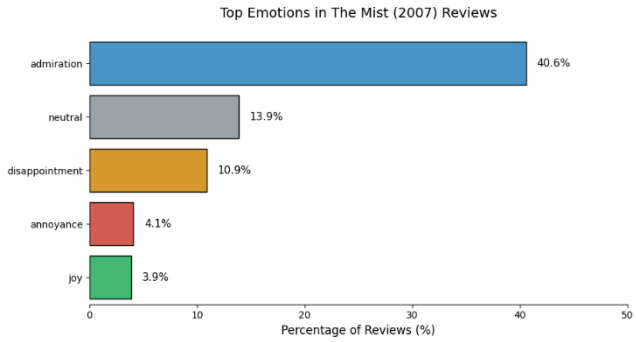

Figure 2

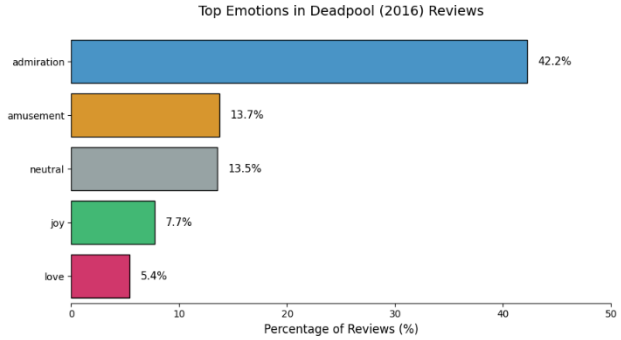

Figure 3

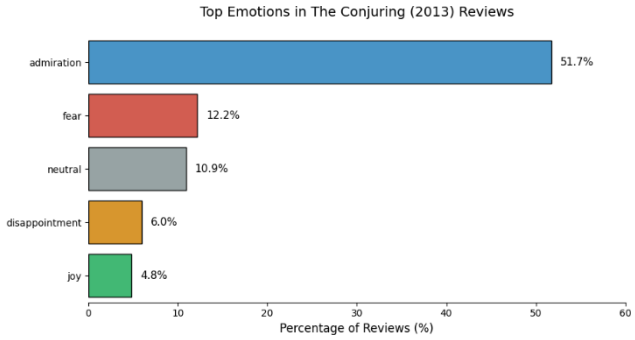

Figure 4

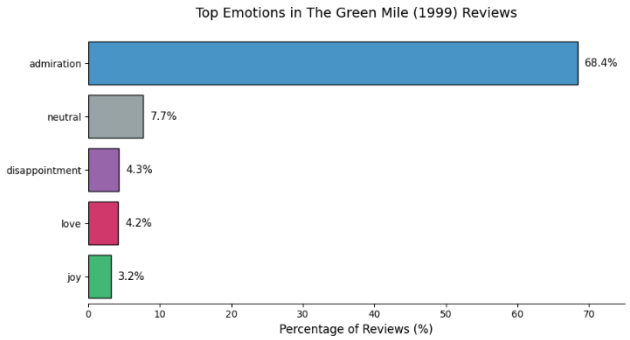

Figure 5

Emotion analysis across all the selected movies share the trend of *admiration* being the dominant response (40.6–68.4%), which could generally depict the audience's praise and appreciation for movies under any genre or category. However, secondary emotions emerge revealing genre-specific audience responses. For example, comedies like Deadpool demonstrate *amusement* (13.7%) while horror films like The Conjuring highlight measurable *fear* (12.2%).

5.2. Cross-Genre Sentiment and Emotion Analysis

This subsection deals with RQ2: Are different emotions (joy, fear, anger, etc.) dominant across different movie genres?

Sentiment analysis across genres showed different patterns as distributed in Figure 6. Reviewers showed strong and dominant positive sentiment toward genres such as Romance (60.4%), Drama (61.1%), and Comedy (61.2%), indicating their enthusiasm for emotionally engaging and lighthearted narratives. The thriller and horror genres, on the other hand, revealed greater negative sentiment percentages, with 26.0% and 37.8%, respectively. These findings suggest that suspenseful storytelling and darker themes create a wide range of emotional responses among audiences.

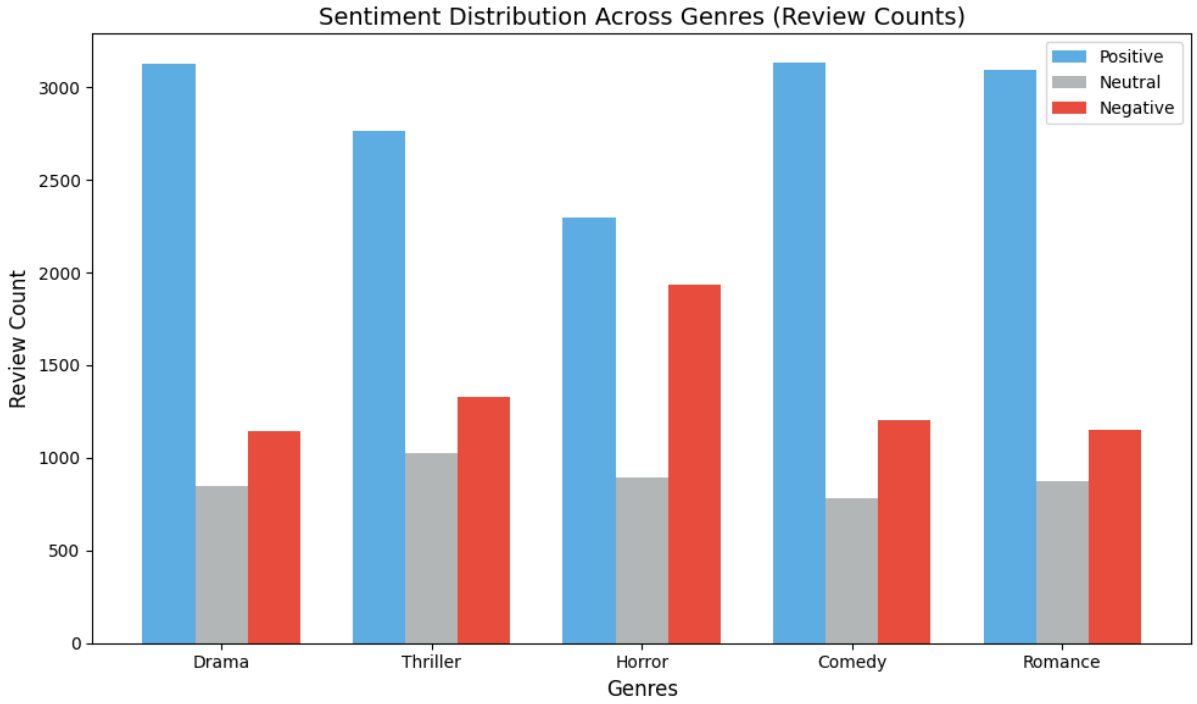

Figure 6

Emotion detection uncovered furhter genre-based user reactions. Similarily, admiration was the most frequent emotion across all five genres, ranging from 53.9% in Horror to 68.2% in Drama. Interesting patterns were emphasized by secondary emotions, like amusement was particularly high in comedy (9.8%). The following figures further illustrate these observations.

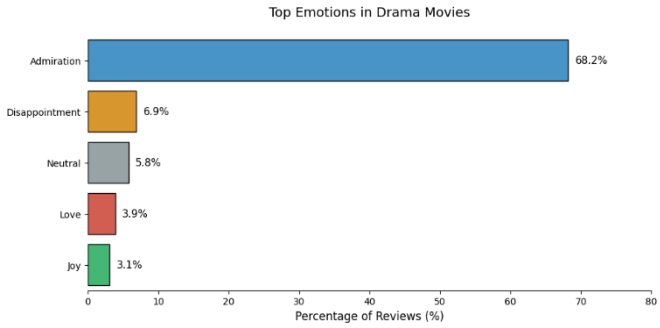

Figure 7

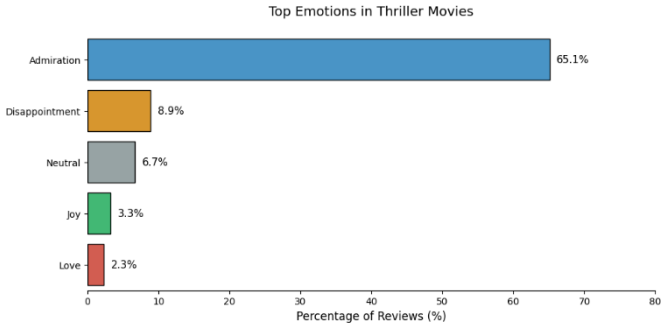

Figure 8

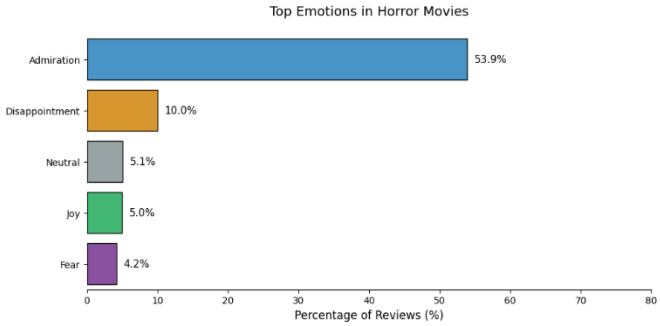

Figure 9

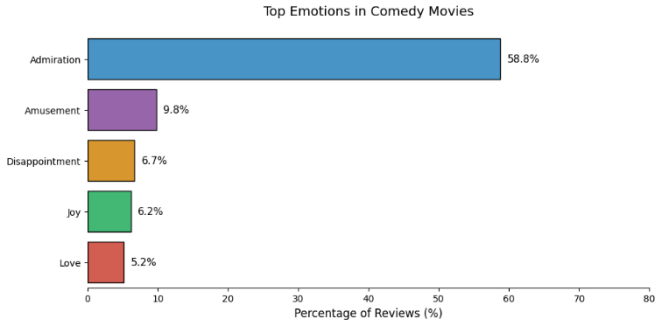

Figure 10

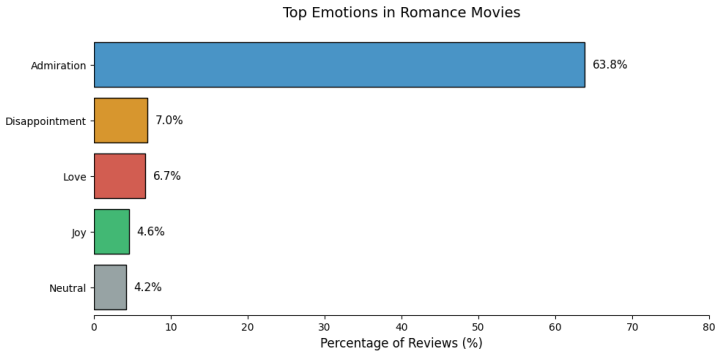

Figure 11

6. Conclusion

This research paper performed sentiment analysis and emotion detection at the individual movie and genre-specific levels using RoBERTa transformer-based models on a dataset of IMDb movie reviews. Sentiment analysis revealed that movie genres like comedy, romance, and drama primarily received favorable sentiments, while horror and thriller tended to be more negative, illustrating the diverse range of reviewers responses impacted by movie themes and narrative styles. The research demonstrated admiration as the leading emotion which prevailed throughout all movie types and genres while amusement in comedies and fear in horror movies represented secondary emotions for genre-specific audience reactions. In this sense, it was also found that in some cases higher movie rating scores do not necessarily match audience's preferences expressed in user reviews.

This study endured several challenges and limitations, including the restriction on analyzing only English-language reviews, rather than sufficient application of sentiment to a more multilingual audience. The changing nature of language creates difficulties for identifying subtle aspects in user reviews because slang and sarcasm and new expressions emerge. Furthermore, The large dataset processing during batch inference faced computational challenges because of speed and memory constraints which demonstrated the requirement for better efficiency methods.

Future research should overcome these restrictions by extending the analysis to multilingual datasets to reach a wider range of audience members. Research into how audience sentiments and emotional responses change across different time periods of movie releases could uncover additional patterns of interest. The analysis of movie emotional responses would benefit from topic modeling techniques to understand which specific aspects of movies create certain emotional reactions.

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